Real Data Implementation Plan

IBM BOB build brief for a real-data, local-run CoolRoute app

> **Implementation target:** a professional local-first app that uses real open street/weather data, provides both the shortest and cooler route, and reports where its shade information is weak.

1. Product acceptance checklist

- Map view uses real OpenStreetMap map tiles with visible contributor credit.
- Users can set start and destination by map click or browser location; map clicks snap to the nearest street node.
- The UI shows distance, estimated walk time, shade share, heat comparison, budget status, and data warnings for both routes.
- Time, heat budget, step-free, shade use, street closure, swap, reset, recenter and theme controls work.
- Mobile and desktop layouts retain legible type, keyboard focus, large controls and consistent action colors.
- Every estimate is labeled accurately; estimated shade is not presented as measured shade, and heat is not presented as medical guidance.

2. Data-to-route design

Stage	Implementation
Load	Overpass returns a bounded set of walkable OSM ways, trees and building footprints.
Prepare	Split ways into graph edges, measure segment lengths and infer a low-confidence shade proxy from nearby mapped objects.
Weather	Request the selected hour from Open-Meteo and use shortwave radiation to scale the relative sun component.
Route	Run shortest-path and heat-budget searches locally; apply closures and step-free filters before search.
Explain	Return route coordinates, distance, walk time, shaded share, confidence, warnings and provider metadata.
Render	Draw the routes on a Leaflet map and show plain-language trade-offs with the limits beside the result.

3. API and operational rules

- No key is embedded in the browser; the local backend calls external services.
- Keep the map window small, cache street responses, use request timeouts, and show useful errors when a provider is down.
- Follow OSM tile caching and attribution rules; do not bulk download or prefetch tiles.
- Use Open-Meteo's free API only for non-commercial use; buy a suitable plan or replace the provider before commercial use.
- Public Overpass and tile servers have no uptime commitment. Production requires a service plan or self-hosting.
- Avoid public Nominatim generic geocoding in this vibe-coded app. Map click and browser location avoid that dependency.

4. Local setup for IBM BOB

```
Open Project/ as the workspace root
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m uvicorn api_server:app --host 127.0.0.1 --port 8765
Browse to http://127.0.0.1:8765
```

5. Future accuracy work

Priority	Upgrade	Evidence needed
1	Validate OSM graph connectivity and tag handling in a few real neighborhoods.	Route walk-through notes and snapped-point error.
2	Replace tree/building proximity with solar geometry, canopy/building height and dated imagery where licensed.	Shade percent observations by time and street side.
3	Add persistent cache, retries, circuit-breaker status and configurable providers.	Outage and rate-limit behavior.
4	Select commercial data, map and weather providers or self-host the needed services.	Provider terms, costs and usage forecasts.
5	Add accessibility field checks and a privacy review before deployment.	Curb and crossing audit plus data-flow review.